

Functions of Biological Molecules

Function of Carbohydrates

- substrate for respiration to provide energy for cell activities(immediate source of energy)
 - Formation of nucleic acids - DNA
 - form supporting structures - cell walls in plants
 - converted to other organic compounds like proteins and fats
 - synthesize lubricants $\Rightarrow$ mucus $\Rightarrow$ which is made up of a carbohydrate and a protein
 - synthesize nectar in some flowers
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Functions of Proteins

- Synthesis of new cytoplasm for growth and repair of worn out body cells
 - Synthesizes enzymes and some hormones
 - formation of antibodies to combat diseases*
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Functions of Fats

- source and long term storage form of energy
 - suitable for long term storage as fats have higher energy value compared to proteins and carbohydrates (2 times more)
- Insulating materials that reduce excessive heat loss
 - animals, example seal $\Rightarrow$ have a thick layer of fat beneath their skin called blubber to reduce loss of body heat
- Solvent for fat soluble vitamins and other vital substances like hormones
- essential part of cells, especially cell membranes
- way to reduce water loss from skin surface; glands in skin secrete an oily substance
 - oily substance forms a thin layer over the skin surface $\rightarrow$ reduce water loss by evaporation
 - Oily substance also reduces rate of heat loss from the skin

